PHYS 142/242 — Assignment 2 & Oral Checkpoint 2 Study Guide

Tunneling in the double well ($\alpha = 0.3282$)

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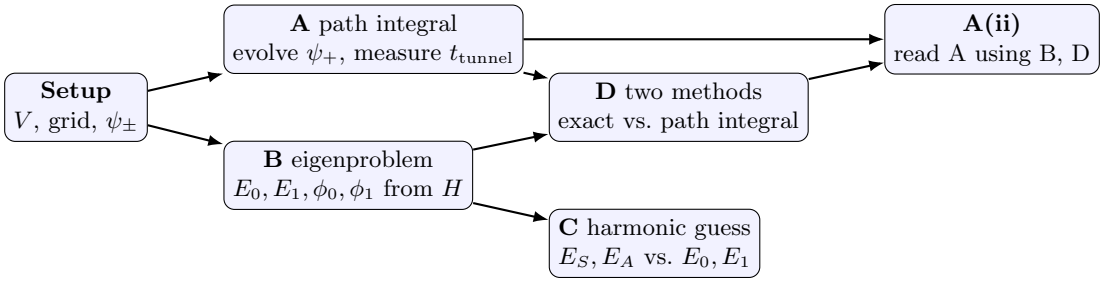
Part I — Understanding the homework

1 The big picture

A particle of mass $m = 1$ ($\hbar = 1$) lives in the symmetric double well

$$V(x) = \alpha x^4 - 2x^2 + \frac{1}{\alpha}, \qquad \alpha = 0.3282.$$

It starts in the right well with energy $E \approx 1.3$, below the barrier $V(0) = 1/\alpha \approx 3.05$. Classically it is trapped; quantum mechanically it *tunnels* to the left well and back. The four problems attack this from different sides:



- **A** evolves the wave function in time with the short-time propagator (no eigenstates needed) and measures how long tunneling takes.
- **B** solves the time-independent problem with a matrix: the two lowest states form a nearly degenerate doublet whose gap ΔE sets the tunneling time $\pi/\Delta E$.
- **C** asks how good the cheap "two Gaussians" picture is: good for the energies, bad for the gap.
- **D** compares the exact two-state evolution with A's path-integral code, which isolates the *numerical* error of the path integral.

Numerical conventions (from the assignment). $N_D + 1 = 601$ points on $[-4, 4]$, $\Delta x = 8/600$; time step $\epsilon = \pi/128$; every integral is $\int f\,dx \approx \Delta x \sum_{i=0}^{N_D} f(x_i)$, *including both endpoints.*

2 Setup: the potential and the harmonic approximation

Minima and barrier.

$$V'(x) = 4\alpha x^3 - 4x = 0 \;\Rightarrow\; x = 0 \text{ (barrier top)} \quad \text{or} \quad x = \pm x_{\min} = \pm\frac{1}{\sqrt{\alpha}} = \pm 1.7455,$$

$$V(\pm x_{\min}) = \frac{\alpha}{\alpha^2} - \frac{2}{\alpha} + \frac{1}{\alpha} = 0, \qquad V(0) = \frac{1}{\alpha}.$$

Curvature, independent of α.

$$V''(x) = 12\alpha x^2 - 4 \;\Rightarrow\; V''(x_{\min}) = 12\alpha \cdot \frac{1}{\alpha} - 4 = 8 \;\Rightarrow\; V_\pm(x) = 4(x \mp x_{\min})^2 = \tfrac{1}{2}\omega^2(x \mp x_{\min})^2, \quad \omega = 2\sqrt{2}.$$

Local ground states and their combinations.

$$\psi_\pm(x) = \left(\frac{\omega}{\pi}\right)^{1/4} e^{-\frac{\omega}{2}(x \mp x_{\min})^2}, \qquad \psi_S = \frac{\psi_+ + \psi_-}{\sqrt{2}} \text{ (even)}, \qquad \psi_A = \frac{\psi_+ - \psi_-}{\sqrt{2}} \text{ (odd)}.$$

$\psi_\pm$ is a minimum-uncertainty Gaussian: $\Delta x = 1/\sqrt{2\omega} = 0.42$, $\Delta p = \sqrt{\omega/2} = 1.19$, $\Delta x \Delta p = \frac{1}{2}$. These do not depend on α; only the position $x_{\min}$ does.

Exact expansion around a minimum (used in A and C). With $u = x - x_{\min}$ and $\alpha x_{\min}^2 = 1$, the constant and linear terms cancel:

$$V(x) = \underbrace{4u^2}_{V_+} + 4\sqrt{\alpha}\, u^3 + \alpha u^4.$$

The cubic term is negative on the inner side ($u < 0$), so the true barrier is *softer* than the parabola. On the outer side ($u > 0$) the wall is *steeper*.

```
TSN = "200288141"
ALPHA = 0.3 + int(TSN[-3:]) / 5000 # = 0.3282
BOXSIZE, ND = 8.0, 600 # interval [-4, 4], 600 intervals
DELTAX = BOXSIZE / ND
x = np.linspace(-BOXSIZE / 2, BOXSIZE / 2, ND + 1) # 601 points, both endpoints included
DELTAT = np.pi / 128 # time step epsilon
XMIN = 1 / np.sqrt(ALPHA) # well minima at +/- XMIN
OMEGA = 2 * np.sqrt(2) # harmonic frequency (V'' = 8 = omega^2)
N_EXPORT = np.arange(0, 2049, 64) # results.json steps n = 0, 64, ..., 2048
```

```
def integrate(f):
    """Grid integral: dx * sum over all 601 points (assignment convention)."""
    return DELTAX * np.sum(f)
```

`integrate` is the single implementation of $\int f\,dx$; every normalization, expectation value and overlap goes through it, so all problems use the same convention.

3 Problem A: path-integral tunneling

3.1 Propagator

For a time-independent $\hat{H}$,

$$\psi(x_b, t+\epsilon) = \int K(x_b, \epsilon; x_a, 0)\, \psi(x_a, t)\, dx_a, \qquad K(x_b, \epsilon; x_a, 0) = \langle x_b | e^{-i\hat{H}\epsilon} | x_a \rangle.$$

K depends only on the step length ϵ, never on t, so one kernel advances the state at every step.

3.2 Short-time kernel

Over a short step the dominant path is the straight line with velocity $v = (x_b - x_a)/\epsilon$. Its action, with the midpoint rule for $\int V\,dt$, is

$$S = \int_0^\epsilon \left[\tfrac{1}{2}\dot{x}^2 - V(x)\right] dt \approx \frac{(x_b - x_a)^2}{2\epsilon} - \epsilon\, V\!\left(\frac{x_a + x_b}{2}\right), \qquad K \approx A\, e^{iS}, \quad A = \frac{1}{\sqrt{2\pi i \epsilon}}.$$

Dropping A gives the assignment's kernel

$$\boxed{\tilde{K}(x_b, \epsilon; x_a, 0) = \exp\Big[i\Big(\frac{(x_b - x_a)^2}{2\epsilon} - V\Big(\frac{x_a + x_b}{2}\Big)\epsilon\Big)\Big]}$$

(first term: free-particle phase; second: phase picked up from the potential during the step).

3.3 Why the prefactor can be dropped

A is a constant (independent of x_a, x_b, t). Dropping it multiplies ψ by the same factor everywhere, which leaves its shape and relative phases unchanged, and the renormalization

$$\psi \leftarrow \frac{\psi}{\sqrt{\int |\psi|^2 dx}}$$

after every step removes the factor. Numerically, each un-renormalized step shrinks ψ by $|1/A| = \sqrt{2\pi\epsilon} = 0.39$ (checked), so without renormalizing ψ would underflow to 0 within ~ 760 steps. Renormalizing also absorbs the small non-unitarity of a finite grid.

3.4 From the integral to a matrix

On the grid the integral becomes a Riemann sum, i.e. a matrix–vector product:

$$\psi_i(t + \epsilon) = \Delta x \sum_{j=0}^{N_D} \tilde{K}_{ij}\, \psi_j(t), \qquad \tilde{K}_{ij} = \tilde{K}(x_i, \epsilon; x_j, 0),$$

where row i is the arrival point x_b and column j the starting point x_a. Using $x_i - x_j = (i-j)\Delta x$ and $\frac{x_i + x_j}{2} = x_{(i+j)/2}$, every entry factorizes:

$$\tilde{K}_{ij} = p_{|i-j|}\, q_{i+j}, \qquad p_k = e^{i\kappa k^2} \text{ (kinetic)}, \quad q_s = e^{-i\epsilon V(x_{s/2})} \text{ (potential)}, \quad \kappa = \frac{\Delta x^2}{2\epsilon} \approx 3.62 \times 10^{-3}.$$

$$\tilde{K} = \begin{pmatrix} q_0 & p_1 q_1 & p_2 q_2 & \cdots & p_{N_D} q_{N_D} \\ p_1 q_1 & q_2 & p_1 q_3 & \cdots & p_{N_D-1} q_{N_D+1} \\ p_2 q_2 & p_1 q_3 & q_4 & \cdots & p_{N_D-2} q_{N_D+2} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ p_{N_D} q_{N_D} & p_{N_D-1} q_{N_D+1} & p_{N_D-2} q_{N_D+2} & \cdots & q_{2N_D} \end{pmatrix}.$$

For $N_D = 2$ (the top-left corner of the same pattern):

$$\tilde{K} = \begin{pmatrix} e^{-i\epsilon V(x_0)} & e^{i\kappa} e^{-i\epsilon V(x_{1/2})} & e^{4i\kappa} e^{-i\epsilon V(x_1)} \\ e^{i\kappa} e^{-i\epsilon V(x_{1/2})} & e^{-i\epsilon V(x_1)} & e^{i\kappa} e^{-i\epsilon V(x_{3/2})} \\ e^{4i\kappa} e^{-i\epsilon V(x_1)} & e^{i\kappa} e^{-i\epsilon V(x_{3/2})} & e^{-i\epsilon V(x_2)} \end{pmatrix}.$$

Properties.

- **Symmetric** ($\tilde{K} = \tilde{K}^T$): $(i-j)^2$ and $i+j$ are unchanged by swapping. It is complex symmetric, *not* Hermitian.
- **Dense, pure phases:** $|\tilde{K}_{ij}| = 1$. In one step every point sends amplitude to every point.
- The **kinetic** factor is constant along diagonals (it depends only on the jump length and not on α); the **potential** factor is constant along anti-diagonals (it depends on the midpoint, and is the only place α enters). The diagonal is $e^{-i\epsilon V(x_i)}$: staying put costs only potential phase.
- **Effective band:** the kinetic phase stays below π for $|i-j| \lesssim \sqrt{\pi/\kappa} \approx 29$ points ($\sqrt{2\pi\epsilon} \approx 0.39$ in x). Longer jumps spin rapidly and cancel in the sum, so one step really moves amplitude only ~ 0.4.
- **Resolution limit:** neighbouring entries in a row differ in phase by $\kappa(2k+1)$, $k = |i-j|$. Beyond $k \approx 433$ this exceeds π (under-resolved). At $k^* = \pi/\kappa \approx 867$ it reaches 2π and the sum would behave like a fake stationary point (a "ghost path"). That is harmless here because the longest jump is $N_D = 600 < 867$ (see Problem D).
- $\tilde{K} \approx e^{-iH\epsilon}$ is dense although H (Problem B) is tridiagonal, because exponentiating a local operator couples every point.

3.5 Algorithm and code

$$\boldsymbol{\phi} = \Delta x\, \tilde{K}\boldsymbol{\psi}^{(n)}, \qquad \boldsymbol{\psi}^{(n+1)} = \frac{\boldsymbol{\phi}}{\sqrt{\Delta x \sum_i |\phi_i|^2}}, \qquad \langle x\rangle(n\epsilon) = \Delta x \sum_i x_i\, |\psi_i^{(n)}|^2.$$

```
1  @njit
2  def build_kernel(x, dt, alpha):
3      """K~_ij = exp{ i[ (x_i - x_j)^2 / (2 dt) - V((x_i + x_j)/2) dt ] }; row i = arrival, column j = start."""
4      K = np.empty((x.size, x.size), dtype=np.complex128)
5      for i in range(x.size):
6          for j in range(x.size):
7              K[i, j] = np.exp(1j * (0.5 * (x[i] - x[j])**2 / dt - V(0.5 * (x[i] + x[j]), alpha) * dt))
8      return K
9
10 @njit
11 def evolve(psi0, K, dx, nt):
12     psi = psi0.astype(np.complex128) / np.sqrt(dx * np.sum(np.abs(psi0)**2))
13     prob = np.empty((nt + 1, psi.size))
14     prob[0] = np.abs(psi)**2
15     for n in range(nt):
16         psi = dx * (K @ psi) # one step: matrix-vector product
17         psi = psi / np.sqrt(dx * np.sum(np.abs(psi)**2)) # renormalize (prefactor dropped)
18         prob[n + 1] = np.abs(psi)**2
19     return prob
20
21 K_tilde = build_kernel(x, DELTAT, ALPHA) # built once (first call also compiles)
22 T_RUN = 48 * np.pi # >= 16 pi and >= 2 pi/dE (checked in B)
23 t = np.arange(int(np.ceil(T_RUN / DELTAT)) + 1) * DELTAT
24 x_mean_A = DELTAX * (evolve(psi_plus, K_tilde, DELTAX, t.size - 1) @ x) # <x>(t) = dx sum_i x_i |psi_i|^2
25 n_tunnel = np.argmin(x_mean_A) # deepest point = first major minimum
26 t_tunnel = t[n_tunnel]
27 x_mean_pi = x_mean_A[N_EXPORT] # for results.json
```

Math	Code
$\tilde{K}_{ij}$ (boxed kernel)	line 7: `K[i, j] = np.exp(1j * (...))`
$\Delta x \sum_j \tilde{K}_{ij}\psi_j$	line 16: `dx * (K @ psi)`
$\psi \leftarrow \psi/\sqrt{\Delta x \sum \lvert\psi\rvert^2}$	line 17 (and line 12 for the initial state)
$\langle x\rangle = \Delta x \sum_i x_i \lvert\psi_i\rvert^2$	line 24: `DELTAX * (prob @ x)` (all time steps at once)
t_{tunnel} = first major minimum	lines 25–26: `argmin` over a run of 48π

`@njit` compiles the double loop ($601^2 \approx 3.6 \times 10^5$ complex exponentials) and the 6144 matrix–vector products. `V` must also be `@njit` because the compiled loop calls it.

3.6 Run length and the tunneling time

The run must cover $\geq 16\pi$ (for `results.json`) and at least one slow period $2\pi/\Delta E \approx 132$. I use $48\pi \approx 151$ (6144 steps). Since $2\pi/\Delta E < 48\pi < 3\pi/\Delta E \approx 198$, the run contains exactly one major minimum, so the global `argmin` is the first major minimum. Zero crossings are unreliable, because the fast wiggles cross zero several times.

3.7 What to expect: the eigenstate picture

Expanding the initial state in eigenstates, $\psi_+ = \sum_n c_n \phi_n$ ($c_n = \int \phi_n \psi_+ dx$, real), each component only acquires a phase:

$$\psi(x,t) = \sum_n c_n e^{-iE_n t}\phi_n \;\Rightarrow\; \langle x\rangle(t) = \sum_{m,n} c_m c_n\, x_{mn} \cos\big((E_n - E_m)t\big), \qquad x_{mn} = \int \phi_m\, x\, \phi_n\, dx.$$

- **Parity:** x is odd, so $x_{mn} \neq 0$ only for opposite-parity pairs.
- **Slow transfer:** 94.7% of ψ_+ is in the doublet ϕ_0, ϕ_1, which gives $\propto \cos(\Delta E\, t)$ with period $2\pi/\Delta E$: right $\to$ left $\to$ right.
- **Fast wiggles:** the other 5.3% is in higher states, which beat with the doublet at the next level spacing $E_2 - E_{0,1} \approx 1.8$ (period ≈ 3.4). Physically the Gaussian is not exactly the well's ground state (the true well is asymmetric), so the packet sloshes inside the well.

Results (A)

$t_{\text{tunnel}} = 63.96$, $\langle x \rangle = -1.717$, compared with $\pi/\Delta E = 65.84$ and the two-state value $-x_{01} = -1.550$. The minimum is earlier and deeper because a fast low point coincides with the slow minimum. D's pure two-state run, with the same code, has no such wiggles, so the shift is physical (higher states) and not numerical. The fast period is ≈ 3.4 (median spacing of local minima).

4 Problem B: numerical eigenstates

4.1 Discretizing the Schrödinger equation

We need $-\frac{1}{2}\phi'' + V\phi = E\phi$. Taylor-expand around x_i:

$$\phi_{i\pm1} = \phi_i \pm \Delta x\, \phi_i' + \tfrac{\Delta x^2}{2}\phi_i'' \pm \tfrac{\Delta x^3}{6}\phi_i''' + \tfrac{\Delta x^4}{24}\phi_i'''' + \dots \;\Rightarrow\; \frac{\phi_{i+1} - 2\phi_i + \phi_{i-1}}{\Delta x^2} = \phi_i'' + \frac{\Delta x^2}{12}\phi_i'''' + \mathcal{O}(\Delta x^4).$$

The odd terms cancel, so the centred three-point formula is **second order**. At each grid point:

$$-\frac{1}{2\Delta x^2}\big(\phi_{i+1} - 2\phi_i + \phi_{i-1}\big) + V(x_i)\phi_i = E\phi_i.$$

4.2 The Hamiltonian matrix

With $d_i = \frac{1}{\Delta x^2} + V(x_i)$ and $\tau = \frac{1}{2\Delta x^2}$ (on our grid $1/\Delta x^2 = 5625$, $\tau = 2812.5$), this is $H\boldsymbol{\phi} = E\boldsymbol{\phi}$:

$$H = \begin{pmatrix} d_0 & -\tau & & & \\ -\tau & d_1 & -\tau & & \\ & \ddots & \ddots & \ddots & \\ & & -\tau & d_{N_D-1} & -\tau \\ & & & -\tau & d_{N_D} \end{pmatrix} = \underbrace{\frac{1}{2\Delta x^2}\begin{pmatrix} 2 & -1 & & & \\ -1 & 2 & -1 & & \\ & \ddots & \ddots & \ddots & \\ & & & -1 & 2 \end{pmatrix}}_{T = -\frac{1}{2}\frac{d^2}{dx^2}} + \underbrace{\begin{pmatrix} V(x_0) & & & & \\ & V(x_1) & & & \\ & & \ddots & & \\ & & & & V(x_{N_D}) \end{pmatrix}}_{\text{diag}(V)}.$$

Toy example ($N_D = 4$, so $\Delta x = 2$, $x = -4, -2, 0, 2, 4$, $\alpha = 0.3282$; far too coarse, but it shows the structure):

$$H = \begin{pmatrix} 55.3161 & -0.125 & 0 & 0 & 0 \\ -0.125 & 0.5481 & -0.125 & 0 & 0 \\ 0 & -0.125 & 3.2969 & -0.125 & 0 \\ 0 & 0 & -0.125 & 0.5481 & -0.125 \\ 0 & 0 & 0 & -0.125 & 55.3161 \end{pmatrix}$$

The large corners are the outer walls ($V(\pm4) \approx 55$), the centre is the barrier ($1/\alpha$), and the small entries at $x = \pm2$ are near the wells.

4.3 Boundary conditions

Row 0 has no left neighbour:

$$d_0\phi_0 - \tau\phi_1 = E\phi_0 \quad\Longleftrightarrow\quad -\frac{1}{2\Delta x^2}\big(\phi_1 - 2\phi_0 + \underbrace{\phi_{-1}}_{=0}\big) + V(x_0)\phi_0 = E\phi_0.$$

So the matrix imposes $\phi = 0$ at the ghost points $x_0 - \Delta x$ and $x_{N_D} + \Delta x$: hard walls just *outside* the box.

4.4 Properties of H

- **Real symmetric:** the eigenvalues are real and the eigenvectors are real and orthogonal.
- **Sparse:** $3N_D + 1 \approx 1800$ nonzeros out of $601^2 \approx 361{,}000$, so it is stored with `scipy.sparse`.
- **Parity:** with the reversal matrix J (ones on the anti-diagonal; e.g. for 3×3, $J = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{pmatrix}$), $JHJ = H$ because $V(-x) = V(x)$. So every non-degenerate eigenvector satisfies $J\boldsymbol{\phi} = \pm\boldsymbol{\phi}$: ϕ_0 is even (no node) and ϕ_1 odd (node at 0).
- **Positive definite:** T has eigenvalues $\frac{2}{\Delta x^2}\sin^2\frac{k\pi}{2(N_D+2)} > 0$ and $V \geq 0$, so all $E_n > 0$.
- **Spectrum:** $0 < E_n \lesssim \frac{2}{\Delta x^2} + \max V \approx 11{,}300$ (Gershgorin). The states we need (~ 1.3) sit at the very bottom.

4.5 The solver: shift-invert and inverse iteration

Krylov methods (ARPACK, behind `eigsh`) converge fastest to eigenvalues of *largest* magnitude. Shift-invert maps the ones we want there:

$$H\boldsymbol{\phi}_n = E_n\boldsymbol{\phi}_n \iff (H-\sigma I)^{-1}\boldsymbol{\phi}_n = \frac{1}{E_n-\sigma}\boldsymbol{\phi}_n.$$

The eigenvectors are unchanged; the E_n nearest σ become the largest. $\sigma = 0$ is valid because all $E_n > 0$. Harmonic estimates ($\omega/2 \approx 1.4$ for the doublet, $3\omega/2 \approx 4.2$ next) give $1/E \approx 0.7$ vs. 0.24: well separated, so convergence takes only a few iterations.

Inverse power iteration (the idea): repeat $(H-\sigma I)\boldsymbol{u}_{k+1} = \boldsymbol{u}_k$, $\boldsymbol{u}_{k+1} \leftarrow \boldsymbol{u}_{k+1}/\|\boldsymbol{u}_{k+1}\|$. The component along ϕ_n grows like $(E_n-\sigma)^{-k}$, so the state nearest σ wins.

Never form the inverse (it is dense). Factor the tridiagonal matrix once, $H - \sigma I = LU$:

$$L = \begin{pmatrix} 1 & & & \\ \ell_1 & 1 & & \\ & \ddots & \ddots & \\ & & \ell_{N_D} & 1 \end{pmatrix}, \quad U = \begin{pmatrix} u_0 & -\tau & & \\ & u_1 & -\tau & \\ & & \ddots & \ddots \\ & & & u_{N_D} \end{pmatrix}, \quad u_0 = d_0 - \sigma, \quad \ell_i = -\frac{\tau}{u_{i-1}}, \quad u_i = d_i - \sigma - \frac{\tau^2}{u_{i-1}}$$

(from matching $(LU)_{i,i-1} = \ell_i u_{i-1} = -\tau$ and $(LU)_{ii} = u_i - \tau\ell_i = d_i - \sigma$). Each iteration is then a forward and a back substitution, costing $\mathcal{O}(N_D)$.

4.6 From eigenvectors to wave functions

`eigsh` returns unit vectors, $\sum_i |v_i|^2 = 1$, but a wave function needs $\Delta x \sum_i |\phi_i|^2 = 1$:

$$\phi_i = \frac{v_i}{\sqrt{\Delta x}} \quad \Rightarrow \quad \Delta x \sum_i |\phi_i|^2 = \Delta x \cdot \frac{1}{\Delta x}\sum_i |v_i|^2 = 1.$$

Signs are arbitrary (ARPACK starts from a random vector), so I fix $\phi_0 > 0$ and the right lobe of $\phi_1 > 0$. Then $(\phi_0 + \phi_1)/\sqrt{2}$ lies in the *right* well and $x_{01} > 0$.

4.7 Expected accuracy (B(i), B(ii))

- **Grid error.** The discrete kinetic operator is $T_d = T - \frac{\Delta x^2}{24}\frac{d^4}{dx^4} = T - \frac{\Delta x^2}{24}p^4$. To first order, $\delta E_n \approx -\frac{\Delta x^2}{24}\langle p^4\rangle_n$. For a harmonic ground state, p is Gaussian with $\langle p^2\rangle = \omega/2$, so $\langle p^4\rangle = 3(\omega/2)^2 = 6$ and

$$\delta E \approx -\frac{(8/600)^2}{24}\cdot 6 \approx -4\times 10^{-5} \qquad \text{(too low; 4–5 matching digits).}$$

- **Box error.** The walls push E up, but $V(\pm 4) \approx 55 \gg E$, so it is negligible.
- **Gap.** E_0 and E_1 have similar same-sign errors, which cancel in ΔE.
- **Richardson.** If $E(\Delta x) = E^* + c\Delta x^2$, halving Δx removes $\frac{3}{4}$ of the error, and $E^* \approx E_{\text{half}} + \frac{1}{3}(E_{\text{half}} - E_{\text{orig}})$.

4.8 Code

```
1  def make_hamiltonian(alpha, L=BOXSIZE, nd=ND):
2      """Sets the grid (dx = L/nd) and builds the sparse three-point Hamiltonian H."""
3      xg = np.linspace(-L / 2, L / 2, nd + 1)
4      dx = L / nd # grid spacing
5      off = np.full(nd, -1 / (2 * dx**2)) # H_{i,i+1} = H_{i+1,i} = -1/(2 dx^2)
6      H = scipy.sparse.diags([off, 1 / dx**2 + V(xg, alpha), off], [-1, 0, 1], format="csc") # H_ii = 1/dx^2 + V
7      return xg, dx, H
8
9  def lowest_two(alpha, L=BOXSIZE, nd=ND):
10     """Two lowest eigenpairs by shift-invert (sigma = 0), normalized and sign-fixed."""
11     xg, dx, H = make_hamiltonian(alpha, L, nd)
12     E, v = scipy.sparse.linalg.eigsh(H, k=2, sigma=0, which="LM")
13     order = np.argsort(E)
14     E, phi = E[order], v[:, order] / np.sqrt(dx) # unit vectors -> dx sum |phi|^2 = 1
15     phi[:, 0] *= np.sign(np.sum(phi[:, 0])) # phi0 positive
16     phi[:, 1] *= np.sign(np.sum(phi[xg > 0, 1])) # right lobe of phi1 positive
17     return xg, dx, H, E[0], E[1], phi[:, 0], phi[:, 1]
```

Math	Code
$\Delta x = L/N_D$ ("the code that sets the grid spacing")	line 4
$-\tau$ off-diagonals, d_i diagonal, tridiagonal sparse H	lines 5–6 (`diags` with offsets $-1, 0, 1$)
boundary rows (no outside neighbour)	automatic: `off` has length N_D, so rows $0, N_D$ get one neighbour
shift-invert, $\sigma = 0$	line 12: `eigsh(H, k=2, sigma=0, which="LM")`
sort, $\phi = v/\sqrt{\Delta x}$	lines 13–14
sign convention	lines 15–16

Results (B)

B(i) (vs. Banerjee & Bhatnagar, $\lambda = \alpha/4$): 4–5 digits agree. All errors are negative, -2 to -4×10^{-5} (as predicted), and halving Δx divides every error by exactly 4.00, so the agreement is limited by the second-order grid error.
B(ii) ($\alpha = 0.3282$): $E_0 = 1.2924391655$, $E_1 = 1.3401570118$, $\Delta E = 0.0477178463$. Half spacing: the energies shift by $+2.6 \times 10^{-5}$ but ΔE only by $+1.4 \times 10^{-6}$ (cancellation). Larger box: $\sim 10^{-9}$. Quoted: $E_0 = 1.2924$, $E_1 = 1.3402$, $\Delta E = 0.04772$.
B(iii): normalized and orthogonal (overlap $\sim 10^{-16}$); ϕ_0 even and nodeless, ϕ_1 odd; nearly identical lobes in each well, hence the near-degeneracy.

5 Problem C: harmonic-approximation energies

5.1 Rayleigh quotient and the variational principle

For any trial $\psi = \sum_n c_n \phi_n$,

$$E[\psi] = \frac{\int \psi^* \hat{H} \psi \, dx}{\int |\psi|^2 dx} = \frac{\sum_n |c_n|^2 E_n}{\sum_n |c_n|^2} \;\geq\; \text{lowest } E_n \text{ present.}$$

ψ_S gives $E_S \geq E_0$. ψ_A is odd, so $c_0 = c_2 = \cdots = 0$ and $E_A \geq E_1$. **Both signed errors must be positive.** At an eigenvector the first-order change of $E[\psi]$ vanishes (the quotient is stationary), so $E[\phi_n + \delta] - E_n = \mathcal{O}(\|\delta\|^2)$: energies are much more accurate than the wave functions.

5.2 Overlap and the denominators

The product of the two Gaussians is a Gaussian at 0:

$$S = \int \psi_+ \psi_- dx = \sqrt{\tfrac{\omega}{\pi}} \int e^{-\frac{\omega}{2}[(x-x_{\min})^2 + (x+x_{\min})^2]} dx = \sqrt{\tfrac{\omega}{\pi}}\, e^{-\omega x_{\min}^2} \int e^{-\omega x^2} dx = e^{-\omega/\alpha} \approx 1.8 \times 10^{-4},$$

$$\int |\psi_S|^2 = \tfrac{1}{2}(1+1+2S) = 1+S, \qquad \int |\psi_A|^2 = 1-S.$$

Dropping the denominators would shift $E_{S,A}$ by $\sim ES \approx 2\times 10^{-4}$: ten times B's grid error and 5% of the harmonic gap.

5.3 On the grid

With B's matrix, $E_S = \boldsymbol{\psi}_S^T H \boldsymbol{\psi}_S / \boldsymbol{\psi}_S^T \boldsymbol{\psi}_S$ (the Δx cancels). Written out:

$$\boldsymbol{\psi}^T H \boldsymbol{\psi} = \underbrace{\frac{1}{2\Delta x^2}\Big[\sum_{i=0}^{N_D-1} (\psi_{i+1}-\psi_i)^2 + \psi_0^2 + \psi_{N_D}^2\Big]}_{\text{kinetic} \approx \frac{1}{2}\int |\psi'|^2/\Delta x} + \underbrace{\sum_i V(x_i)\psi_i^2}_{\text{potential}}.$$

The kinetic part is a sum of squared differences (manifestly positive), and the $\psi_0^2, \psi_{N_D}^2$ terms are the walls.

5.4 Two-state (Rayleigh–Ritz) structure

In the basis $\{\psi_+, \psi_-\}$, with $h_d = \langle\psi_\pm|\hat{H}|\psi_\pm\rangle$ and $h_o = \langle\psi_+|\hat{H}|\psi_-\rangle$ (equal by mirror symmetry), $\mathsf{H}\boldsymbol{c} = E\,\mathsf{S}\boldsymbol{c}$:

$$\begin{pmatrix} h_d & h_o \\ h_o & h_d \end{pmatrix}\begin{pmatrix} 1 \\ \pm 1 \end{pmatrix} = E\begin{pmatrix} 1 & S \\ S & 1 \end{pmatrix}\begin{pmatrix} 1 \\ \pm 1 \end{pmatrix} \;\Rightarrow\; E_S = \frac{h_d + h_o}{1+S}, \quad E_A = \frac{h_d - h_o}{1-S},$$

$$\Delta E_{\text{harm}} = E_A - E_S = \frac{(h_d - h_o)(1+S) - (h_d + h_o)(1-S)}{1-S^2} = \frac{2(Sh_d - h_o)}{1-S^2}.$$

ψ_S, ψ_A are the best combinations of $\psi_\pm$. The gap depends only on S and h_o, i.e. on the **barrier region** where ψ_+ and ψ_- overlap.

5.5 Analytic h_d, h_o and the harmonic gap

Since $\psi_\pm$ is the exact ground state of $-\frac{1}{2}\partial_x^2 + V_\pm$, we have $\hat{H}\psi_\pm = \frac{\omega}{2}\psi_\pm + (V - V_\pm)\psi_\pm$. Gaussian moments ($\sigma^2 = \frac{1}{2\omega}$): $\langle u^3\rangle = 0$, $\langle u^4\rangle = 3\sigma^4$.

$$h_d = \frac{\omega}{2} + \alpha \cdot 3\sigma^4 = \sqrt{2} + \frac{3\alpha}{32}.$$

For h_o: $V - V_- = -4\sqrt{\alpha}\,w^3 + \alpha w^4$ ($w = x + x_{\min}$) and $\psi_+\psi_- = S\times$(normalized Gaussian at 0, variance σ^2). With $a = x_{\min}$, $\langle w^3\rangle = a^3 + 3a\sigma^2$, $\langle w^4\rangle = a^4 + 6a^2\sigma^2 + 3\sigma^4$, $\sqrt{\alpha}a = 1$, $\alpha a^2 = 1$:

$$h_o = S\Big[\frac{\omega}{2} - 4a^2 - 12\sigma^2 + a^2 + 6\sigma^2 + 3\alpha\sigma^4\Big] = S\Big[\frac{\omega}{2} - \frac{3}{\alpha} - \frac{3}{\omega} + \frac{3\alpha}{32}\Big].$$

Then $Sh_d - h_o = 3S(\frac{1}{\alpha} + \frac{1}{\omega})$, so

$$\boxed{\Delta E_{\text{harm}} = \frac{6S}{1-S^2}\Big(\frac{1}{\alpha} + \frac{1}{\omega}\Big)} \approx 3.69\times 10^{-3} \quad \text{vs.} \quad \Delta E = 0.0477.$$

5.6 Why the gap fails while the energies do not

1. **Error arithmetic.** With $\delta_S = E_S - E_0$ and $\delta_A = E_A - E_1$: $\Delta E_{\text{harm}} - \Delta E = \delta_A - \delta_S$. The energy errors are divided by $E \approx 1.3$, the gap error by $\Delta E \approx 0.048$ (27× smaller).
2. **The gap lives at the barrier.** Multiply $-\frac{1}{2}\phi_0'' + V\phi_0 = E_0\phi_0$ by ϕ_1 and vice versa, subtract, integrate over $x > 0$, and use $\phi_1(0) = 0$, $\phi_0'(0) = 0$:

$$-\tfrac{1}{2}\big[\phi_1\phi_0' - \phi_0\phi_1'\big]_0^\infty = (E_0 - E_1)\int_0^\infty \phi_0\phi_1\,dx \;\Rightarrow\; \Delta E = \frac{\phi_0(0)\,\phi_1'(0)}{2\int_0^\infty \phi_0\phi_1 dx} \approx \phi_0(0)\phi_1'(0).$$

3. **The parabola misses the barrier.** $V_+(0) = 4/\alpha = 12.2$ vs. the true $V(0) = 3.05$, so the Gaussian decays far too fast inside the barrier: $\phi_0(0)/\psi_S(0) = 8.6$. Too little amplitude at $x = 0$ means too small a gap.
4. In the **outer walls** the opposite happens ($u^3, u^4 > 0$, steeper wall): $|\phi_0| < |\psi_S|$.

```
psi_S = (psi_plus + psi_minus) / np.sqrt(2) # unnormalized, as defined
psi_A = (psi_plus - psi_minus) / np.sqrt(2)
E_S, E_A = (integrate(p * (H @ p)) / integrate(p * p) for p in (psi_S, psi_A)) # Rayleigh quotients, B's H
S = integrate(psi_plus * psi_minus) # checked against exp(-omega/alpha)
```

Results (C)

$E_S = 1.4431$ ($\delta_S = +0.151$, 11.7%), $E_A = 1.4468$ ($\delta_A = +0.107$, 8.0%); gap 0.00369 vs. 0.0477: 92% too small. $\delta_A - \delta_S = -0.044 \approx -\Delta E$. The grid S and gap match the closed forms to $\sim 10^{-6}$. Log plot: curves agree in the wells; $\phi_0/\psi_S = 8.6$ at $x = 0$ (barrier $|x| < 1.03$).

6 Problem D: comparing two evolution methods

6.1 Initial state

$\psi(x,0) = (\phi_0 + \phi_1)/\sqrt{2}$ with B's signs: the two add on the right and cancel on the left, so the state is in the right well. $\int |\psi|^2 = \frac{1}{2}(1 + 1 + 2\int \phi_0\phi_1) = 1$ by orthonormality. Evolve to the first step at or after $T = 2\pi/\Delta E$ (5365 steps).

6.2 D(i): exact two-state evolution

$$\psi(x,t) = \tfrac{1}{\sqrt{2}}\big(e^{-iE_0 t}\phi_0 + e^{-iE_1 t}\phi_1\big) \;\Rightarrow\; |\psi|^2 = \tfrac{1}{2}\big[\phi_0^2 + \phi_1^2 + 2\phi_0\phi_1\cos(\Delta E\, t)\big] \quad (\phi_n \text{ real}),$$

$$\langle x\rangle(t) = \tfrac{1}{2}\big[x_{00} + x_{11} + 2x_{01}\cos(\Delta E\, t)\big] = \boxed{x_{01}\cos(\Delta E\, t)}, \qquad t_{\text{tunnel}} = \frac{\pi}{\Delta E}.$$

- **Normalization:** the $1/\sqrt{2}$ gives the $\frac{1}{2}$; orthonormality keeps $\int |\psi|^2 = 1$ for all t (the cross term integrates to 0).
- **Symmetry:** ϕ_n^2 even, x odd $\Rightarrow x_{00} = x_{11} = 0$; $\phi_0\phi_1$ odd $\Rightarrow x\phi_0\phi_1$ even $\Rightarrow x_{01} \neq 0$. Exact on the symmetric grid (numerically $\sim 10^{-12}$).
- **Mirror image:** at t_{tunnel}, $|\psi|^2 = \frac{1}{2}(\phi_0 - \phi_1)^2(x) = |\psi(-x,0)|^2$.
- **Probability in each well:** $P_{\text{right}}(t) \approx \cos^2(\Delta E t/2)$ and $P_{\text{left}} \approx \sin^2(\Delta E t/2)$, so $\langle x\rangle \approx x_{01}(P_{\text{right}} - P_{\text{left}})$.

6.3 D(ii)–(iii): the path integral and its error

With only ϕ_0, ϕ_1 present, an exact propagator would reproduce the pure cosine. But $\Delta x\tilde{K}$ only approximates $e^{-iH\epsilon}$; effectively the path integral has slightly different eigenstates and energies:

Pattern in $\langle x\rangle_{\text{PI}} - \langle x\rangle_{\text{eig}}$	**Mechanism**
growing phase drift	$\Delta E_{\text{PI}} \neq \Delta E$: $\cos((\Delta E + \delta)t) - \cos(\Delta E t) \approx -\delta\, t \sin(\Delta E t)$
fast wiggles on the PI curve only	the state leaks into higher PI eigenstates and beats at $\approx E_2 - E_{0,1}$
shrinking amplitude or offset	$\tilde{K}$ is not unitary; renormalization reweights the components

The errors accumulate, so the discrepancy is largest late in the run. **Order:** $\tilde{K}$ is time-symmetric ($\tilde{K}(-\epsilon) = \tilde{K}(\epsilon)^*$, $\tilde{K} = \tilde{K}^T$), so odd orders cancel and the error is $\propto \epsilon^2$.

The ghost-path condition. Along a row the kinetic phase κk^2 advances by $\kappa(2k+1)$ per grid point. When this reaches 2π, at $k^* = \pi/\kappa = 2\pi\epsilon/\Delta x^2$, the sampled phases look stationary again and a spurious path jumping $k^*\Delta x$ in one step contributes coherently. The method needs this outside the box:

$$k^* > N_D = \frac{L}{\Delta x} \iff \boxed{2\pi\epsilon > L\,\Delta x} \qquad (\text{now } 0.154 > 0.107).$$

Halving ϵ alone gives $0.077 < 0.107$: broken. A larger box ($L = 12$) at the same Δx gives $0.160 > 0.154$: also broken (checked: error ~ 1.5). A valid refinement halves ϵ *and* Δx.

```
psi0_D = (phi0 + phi1) / np.sqrt(2) # right-well two-state initial state
T_D = 2 * np.pi / DELTA_E # one full oscillation
t_D = np.arange(int(np.ceil(T_D / DELTAT)) + 1) * DELTAT # up to the first step at or after T
x01 = integrate(phi0 * x * phi1)
x_mean_eig_D = x01 * np.cos(DELTA_E * t_D) # D(i): exact, every step
x_mean_eig = x01 * np.cos(DELTA_E * N_EXPORT * DELTAT) # 33 export times, for results.json
prob_PI_D = evolve(psi0_D, K_tilde, DELTAX, t_D.size - 1) # D(ii): same code as A
x_mean_PI_D = DELTAX * (prob_PI_D @ x)
diff = x_mean_PI_D - x_mean_eig_D
n_max = np.argmax(np.abs(diff[t_D <= T_D])) # eq. (12): only steps with n eps <= T
```

Results (D)

$x_{01} = 1.550$; the PI minimum is at 66.29 (exact 65.84). Max discrepancy 0.075 at $t = 128.7$ ($\langle x\rangle_{\mathrm{PI}} = 1.459$, $\langle x\rangle_{\mathrm{eig}} = 1.535$). Pattern: zero at $t = 0$ and growing, with fast wiggles (period ≈ 3.5) plus a slow negative drift late in the run; at T the PI peak is lower and a small remnant stays in the left well. Check: halving ϵ and Δx together reduces it $4\times$ (to 0.019; verified); halving Δx alone changes nothing.

Part II — Oral Checkpoint 2 preparation

7 Format and how to score 3/3

Five minutes, two questions (same for everyone)

Q1 — "Walk me through your code." Say the stages *in order*, what each does *physically*, point to the cell *without searching*, and give *one non-obvious detail*.
Q2 — "Suppose the assignment changed. . . " Say *exactly* what changes, what *stays the same*, any *dependent quantity*, and *predict* the effect.
Template: *"This stage does X; physically it is Y; it happens here; the subtle point is Z."* Pauses are fine. Follow-ups are neutral.

8 Numbers to know

$x_{\min}$ / $V(0)$	1.7455 / 3.047	ω, Δx, ϵ	2.828, 0.01333, 0.02454
E_0 / E_1 / ΔE	1.29244 / 1.34016 / 0.047718	$\pi/\Delta E$ / $2\pi/\Delta E$	65.84 / 131.67
A: t_{tunnel}, $\langle x\rangle$	63.96, -1.717	two-state weight of ψ_+	94.7%
x_{01}	1.550	fast period	≈ 3.4
E_S / E_A	1.4431 / 1.4468	harmonic gap (error)	0.00369 (92%)
S	1.8×10^{-4}	$\phi_0(0)/\psi_S(0)$	8.6
B(i) digits / error ratio	4–5 / 4.00	D max discrepancy	0.075 at $t = 128.7$

9 Q1 — walk-through script

1. **Setup:** parameters, `V` (`@njit`), `integrate` ($\Delta x \sum$ over all points), `normalize`, $\psi_\pm$, `plot_mean`.
2. **A(i):** `build_kernel` makes $\tilde{K}$ once; `evolve` does $\Delta x \tilde{K}\psi$ and renormalizes; $\langle x\rangle = \Delta x \sum x|\psi|^2$; `argmin` gives t_{tunnel}; export at $n = 0, 64, \ldots, 2048$.
3. **B:** `make_hamiltonian` sets Δx and builds the sparse tridiagonal H; `lowest_two` runs shift-invert `eigsh`, divides by $\sqrt{\Delta x}$ and fixes the signs; B(i) literature check, B(ii) three grids, B(iii) plot.
4. **C:** $\psi_{S,A}$; Rayleigh quotients with B's H (denominators kept); log plot.
5. **D:** $(\phi_0 + \phi_1)/\sqrt{2}$; exact $x_{01}\cos\Delta E t$ vs. the same `evolve`; difference and snapshots.
6. **A(ii)** at the end (needs B and D), then `results.json`.

Non-obvious details (have 2–3 ready)

- Δx **in a step:** $\int K\psi\, dx_a \to \Delta x \sum_j K_{ij}\psi_j$; the same Δx is in the norm and in $\langle x\rangle$.
- **Renormalizing:** the dropped A is a constant; without renormalizing, ψ shrinks by 0.39 per step and underflows in ~ 760 steps.
- $\sqrt{\Delta x}$**:** `eigsh` vectors have $\sum |v|^2 = 1$; wave functions need $\Delta x \sum |\phi|^2 = 1$.
- **Signs:** arbitrary from the solver; fixed so that D's state starts in the right well.
- `sigma=0`**:** H is positive definite, so 0 lies below every E_n.
- `argmin`**:** valid because $2\pi/\Delta E < 48\pi < 3\pi/\Delta E$ (checked in B(ii)).
- **Max discrepancy over** $n\epsilon \le T$ **only:** the last step overshoots T.

10 Q2 — "What if the assignment changed?"

Change	Where in my code / what stays	Prediction
α larger (e.g. 0.5)	Only `ALPHA`; everything follows: $x_{\min}$, $\psi_\pm$, the potential phases of $\tilde{K}$ (anti-diagonals), the diagonal of H. **Dependent:** `T_RUN` $= 48\pi$ is hard-coded (the B(ii) check would print `False`).	Wells move in ($x_{\min} = 1.41$), barrier drops (2.0), $\Delta E = 0.231$, $t_{\text{tunnel}} \approx 13.6$. 48π spans ~ 5 periods, so restrict `argmin` to $t < 3\pi/\Delta E$.
α smaller (0.3)	same	Barrier 3.33, $\Delta E = 0.030$, $t_{\text{tunnel}} \approx 105$; need a longer run ($2\pi/\Delta E \approx 210$).
Start $x_c \neq x_{\min}$	Only the initial state; $\tilde{K}$, H, the grid and C/D stay the same.	Large intra-well sloshing (period ≈ 3.4, set by $E_2 - E_{0,1} \approx 1.8$, not $2\pi/\omega$) on top of the tunneling.
Start at $x_c = 0$	same	Even state: only even eigenstates, $\langle x\rangle \equiv 0$, symmetric spreading, no tunneling oscillation.
Start in left well	same	Mirror image: $-\langle x\rangle(t)$, same t_{tunnel}.
Finer grid	`ND`: Δx, x, $\tilde{K}$, H, $\psi_\pm$; time grid and `N_EXPORT` stay.	B error $\div 4$ per halving; A/D barely change (their error is set by ϵ); cost $\propto N_D^2$.
Coarser grid	same	PI breaks once $N_D \lesssim 415$ ($2\pi\epsilon > L\Delta x$). B just loses accuracy.
Larger box, same Δx	`BOXSIZE` and `ND` together	B: $\sim 10^{-9}$ change. **PI breaks** ($0.160 > 0.154$; error ~ 1.5, checked).
Time step ϵ	`DELTAT`: $\tilde{K}$, step counts. **Dependent:** `N_EXPORT` assumes $\epsilon = \pi/128$; use $n = \text{round}(t_k/\epsilon)$. B and D(i) unchanged.	PI error $\propto \epsilon^2$, but halving ϵ alone breaks the kernel; halve Δx too ($0.075 \to 0.019$).
Initial state ϕ_0	initial vector	Stationary: $\langle x\rangle = 0$, density constant.
Mass m	kinetic phase $m(x_b - x_a)^2/2\epsilon$; kinetic part of $H \propto 1/m$	$\kappa = \sqrt{2m(V-E)}$ grows, so ΔE drops exponentially: much slower tunneling.

Traps worth saying out loud

(1) ϵ cannot be reduced alone ($2\pi\epsilon > L\Delta x$). (2) A larger box at fixed Δx violates the same condition. (3) `T_RUN` and `N_EXPORT` are hard-coded and depend on α and ϵ.

11 Conceptual questions

Q: *If α changes, what happens physically to the wave function, in position and in momentum?*
A: $V''(x_{\min}) = 8$ for every α, so each well holds the same Gaussian ($\Delta x = 0.42$, $\Delta p = 1.19$). α moves the wells ($1/\sqrt{\alpha}$) and sets the barrier ($1/\alpha$). Under the barrier the momentum is imaginary, $\kappa = \sqrt{2(V-E)}$, and $\psi \sim e^{-\int \kappa\, dx}$. Larger α means a lower, thinner barrier, more amplitude at 0, a larger ΔE and faster tunneling:

α	$x_{\min}$	$V(0)$	ΔE	$\pi/\Delta E$	two-state weight of ψ_+
0.30	1.83	3.33	0.030	105.2	95.4%
0.3282	**1.75**	**3.05**	**0.048**	**65.8**	**94.7%**
0.40	1.58	2.50	0.112	27.9	93.0%
0.50	1.41	2.00	0.231	13.6	91.3%
0.60	1.29	1.67	0.359	8.8	90.6%

Larger α also makes the wells more anharmonic ($4\sqrt{\alpha}u^3 + \alpha u^4$), so the Gaussian is a worse start and the fast wiggles are bigger.

Q: *Describe the time evolution physically.*
A: The packet starts trapped in the right well ($E < V(0)$). It is mostly $\phi_0 + \phi_1$, and the relative phase $e^{-i\Delta E t}$ turns it into $\phi_0 - \phi_1$ (left well) at $\pi/\Delta E$ and back at $2\pi/\Delta E$: $P_{\text{right}} \approx \cos^2(\Delta E t/2)$. This is interference of two stationary states, not motion over the barrier. Fast wiggles are the packet sloshing inside a well.

Q: *Why is $t_{tunnel} = 63.96$ and not 65.84?*
A: A fast low point adds to the slow minimum. D's two-state run with the same code gives $\pi/\Delta E$, so the shift is physics, not numerics.

Q: *Why is $\tilde{K}$ dense and H tridiagonal?*
A: $\tilde{K} \approx e^{-iH\epsilon}$; exponentiating a local operator couples all points, though only jumps of ~ 0.4 add coherently.

Q: *Why are your energies slightly too low, and why is ΔE better converged?*
A: The three-point kinetic term is low by $\frac{\Delta x^2}{24}\langle p^4\rangle \approx 4 \times 10^{-5}$; the same-sign errors cancel in the difference.

Q: *Why does the harmonic gap fail?*
A: The gap is set by the amplitude at $x = 0$; the parabola overestimates the barrier $4\times$, so $\psi_S(0)$ is $8.6\times$ too small. The energies are set by the wells, and their error is second order.

Q: *Why do the two methods in D disagree, and how would you test it?*
A: Time-step error of the short-time kernel ($\propto \epsilon^2$). Halve ϵ and Δx together and expect $4\times$ smaller; ϵ alone breaks it.

12 Sanity checks

Name 3 and say what they would show

- Norm $= 1$; $\tilde{K} = \tilde{K}^T$, $|\tilde{K}_{ij}| = 1$; $V(\pm x_{\min}) = 0$, $V'' = 8$; $\langle x\rangle(0) = x_{\min}$.
- Mirror start gives $-\langle x\rangle(t)$; an even start gives $\langle x\rangle \equiv 0$.
- Eigenstates orthonormal, ϕ_0 even/nodeless, ϕ_1 odd; $x_{00} = x_{11} \approx 0$; Rayleigh quotient of ϕ_n equals E_n.
- 4–5 digits vs. the literature; error ratio 4 under $\Delta x/2$; box independence.
- $E_S \geq E_0$, $E_A \geq E_1$; grid S and ΔE_{harm} match the closed forms.
- D: PI vs. $x_{01}\cos\Delta E t$; the density at t_{tunnel} mirrors $t = 0$.
- Limits: $V = 0$ gives free spreading $\sigma^2(t) = \sigma_0^2 + t^2/4\sigma_0^2$; a single harmonic well gives period $2\pi/\omega$; $\langle H\rangle$ should stay constant.

13 Last-minute checklist

- Notebook open, **Restart → Run All** done once (about a minute: Numba compiles first), AI tools closed.
- Locate instantly: Setup → `build_kernel/evolve` → `make_hamiltonian/lowest_two` → Rayleigh quotients → D exact vs. PI → A(ii) → `results.json`.
- Q2 answer = *what changes* + *what stays* + *prediction* (+ dependent quantity: `T_RUN`, `N_EXPORT`, $2\pi\epsilon > L\Delta x$).